DBMS EXPERIMENT 3:

```
-- Create table (correct datatype)
CREATE TABLE employee(
  e_id NUMBER(10),
  e_name VARCHAR2(20),
  address VARCHAR2(20),
  email_id VARCHAR2(20)
);

-- Add column
ALTER TABLE employee
ADD salary NUMBER(10,2) NOT NULL;

-- Create table from another table
CREATE TABLE employee1 AS
SELECT * FROM employee;

-- Create table with selected columns
CREATE TABLE employee2 AS
SELECT e_id, e_name FROM employee;

-- Add column
ALTER TABLE employee2
ADD salary NUMBER(12);

-- Drop column
ALTER TABLE employee2
DROP COLUMN e_name;
```

```
-- Modify column
ALTER TABLE employee2
MODIFY e_id NUMBER(20);

-- Drop table
DROP TABLE employee2;

-- Rename table
RENAME employee1 TO emp1;

-- Remove all records
TRUNCATE TABLE employee;

-- Add primary key with constraint name
ALTER TABLE emp1
ADD CONSTRAINT pri_k PRIMARY KEY (e_id);

-- Drop constraint
ALTER TABLE emp1
DROP CONSTRAINT pri_k;

-- Add primary key (without name)
ALTER TABLE emp1
ADD PRIMARY KEY (e_id);
-- Correct way to drop primary key
ALTER TABLE emp1
DROP PRIMARY KEY;
```

EXPERIMENT NO 04

```
SQL> create table MI(jno number(5), pname varchar2(10));

Table created.

SQL> insert into MI values(45, 'rohit');

1 row created.

SQL> insert into MI values('&jno', '&pname');

Enter value for jno: 93

Enter value for pname: bumrah

old 1: insert into MI values('&jno', '&pname')

new 1: insert into MI values('93', 'bumrah')

1 row created.

SQL> insert into MI values('&jno', '&pname');

Enter value for jno: 63

Enter value for pname: surya

old 1: insert into MI values('&jno', '&pname')

new 1: insert into MI values('63', 'surya')

1 row created.

SQL> desc MI

Name Null? Type

----------------------------------------- -------- ----------------------------

JNO NUMBER(5)

PNAME VARCHAR2(10)

SQL> select *from MI;

JNO PNAME

---------- ----------

45 rohit

93 bumrah

63 surya
```

```
SQL> select *from MI where jno<60;

JNO PNAME

---------- ----------

45 rohit

SQL> select *from MI where jno between 60 and 95;

JNO PNAME

---------- ----------

93 bumrah

63 surya

SQL> select *from MI where pname like 'r%';

JNO PNAME

---------- ----------

45 rohit

SQL> select *from MI where pname like '%h';

JNO PNAME

---------- ----------

93 bumrah

SQL> insert into MI values('&jno', '&pname');

Enter value for jno: 10

Enter value for pname: hardik

old 1: insert into MI values('&jno', '&pname')

new 1: insert into MI values('10', 'hardik')

1 row created.

SQL> insert into MI values('&jno', '&pname');

Enter value for jno: 20

Enter value for pname: tilak

old 1: insert into MI values('&jno', '&pname')

new 1: insert into MI values('20', 'tilak')
```

```
1 row created.

SQL> select *from MI;

JNO PNAME

---------- ----------

45 rohit

93 bumrah

63 surya

10 hardik

20 tilak

SQL> update MI set jno=33 where pname='hardik';

1 row updated.

SQL> update MI set jno=72 where pname='tilak';

1 row updated.

SQL> select *from MI;

JNO PNAME

---------- ----------

45 rohit

93 bumrah

63 surya

33 hardik

72 tilak

SQL> delete from MI where jno=33;

1 row deleted.

SQL> select *from MI;

JNO PNAME

---------- ----------

45 rohit

93 bumrah
```

63 surya

72 tilak

SQL> select *from MI where jno in(40,65);

no rows selected

SQL> select *from MI where jno>=65;

JNO PNAME

---------- ----------

93 bumrah

72 tilak

SQL> select *from MI where jno<=50;

JNO PNAME

---------- ----------

45 rohit

SQL> delete from MI where jno=72;

1 row deleted.

SQL> select *from MI;

JNO PNAME

---------- ----------

45 rohit

93 bumrah

63 surya

Experiment No. 05

Implementation of Aggregate function and Set Operations in DBMS.

SQL*Plus: Release 11.2.0.2.0 Production on Wed Mar 11 09:35:57 2026

Copyright (c) 1982, 2010, Oracle. All rights reserved.

```
SQL> connect
Enter user-name: system
Enter password:
Connected.
SQL> create table dsem1(eid number (20), ename varchar2 (30), sal number (30), dno
number (20));

Table created.

SQL> insert into dsem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 2
Enter value for ename: Varun
Enter value for sal: 50000
Enter value for dno: 6
old   1: insert into dsem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into dsem1 values ('2', 'Varun', '50000', '6')

1 row created.

SQL> insert into dsem1 values ('&eid', '&ename', '&sal', '&dno');
```

```
Enter value for eid: 9
Enter value for ename: Anupam
Enter value for sal: 55000
Enter value for dno: 7
old   1: insert into dsem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into dsem1 values ('9', 'Anupam', '55000', '7')

1 row created.

SQL> insert into dsem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 13
Enter value for ename: Akanksha
Enter value for sal: 50000
Enter value for dno: 18
old   1: insert into dsem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into dsem1 values ('13', 'Akanksha', '50000', '18')

1 row created.

SQL> insert into dsem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 17
Enter value for ename: Om
Enter value for sal: 70000
Enter value for dno: 45
old   1: insert into dsem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into dsem1 values ('17', 'Om', '70000', '45')

1 row created.
```

```
SQL> select *from dsem1;

       EID ENAME                            SAL       DNO
---------- ------------------------------ ---------- ----------
         2 Varun                          50000         6
         9 Anupam                         55000         7
        13 Akanksha                       50000        18
        17 Om                             70000        45

SQL> create table csem1(eid number (20), ename varchar2 (30), sal number (30), dno
number (20));

Table created.

SQL> insert into csem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 1
Enter value for ename: Rohit S
Enter value for sal: 200000
Enter value for dno: 45
old   1: insert into csem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into csem1 values ('1', 'Rohit S', '200000', '45')

1 row created.

SQL> insert into csem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 6
```

```
Enter value for ename: Virat K
Enter value for sal: 15000
Enter value for dno: 18
old   1: insert into csem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into csem1 values ('6', 'Virat K', '15000', '18')

1 row created.

SQL> insert into csem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 8
Enter value for ename: Thala
Enter value for sal: 200000
Enter value for dno: 7
old   1: insert into csem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into csem1 values ('8', 'Thala', '200000', '7')

1 row created.

SQL> select sum(sal) from csem1;

  SUM(SAL)
----------
    415000

SQL> select avg(sal) from dsem1;

  AVG(SAL)
----------
```

```
    56250

SQL> select count(ename) from dsem1;

COUNT(ENAME)
------------
           4

SQL> select *from dsem1;

       EID ENAME                          SAL       DNO
---------- ------------------------------ ---------- ----------
         2 Varun                          50000         6
         9 Anupam                         55000         7
        13 Akanksha                       50000        18
        17 Om                             70000        45

SQL> select *from csem1;

       EID ENAME                          SAL       DNO
---------- ------------------------------ ---------- ----------
         1 Rohit S                       200000        45
         6 Virat K                        15000        18
         8 Thala                         200000         7

SQL> select dno from dsem1 union select dno from csem1;

       DNO
```

```
----------
         6
         7
        18
        45

SQL> select dno from csem1 union select dno from dsem1;

       DNO
----------
         6
         7
        18
        45

SQL> insert into csem1 values ('&eid', '&ename', '&sal', '&dno');
Enter value for eid: 15
Enter value for ename: Tilak
Enter value for sal: 15000
Enter value for dno: 33
old   1: insert into csem1 values ('&eid', '&ename', '&sal', '&dno')
new   1: insert into csem1 values ('15', 'Tilak', '15000', '33')

1 row created.

SQL> select dno from csem1 union select dno from dsem1;

       DNO
```

```
----------
         6
         7
        18
        33
        45

SQL> select dno from csem1 union all select dno from dsem1;

       DNO
----------
        45
        18
         7
        33
         6
         7
        18
        45

8 rows selected.

SQL> select dno from csem1 intersect select dno from dsem1;

       DNO
----------
         7
        18
```

```
        45

SQL> select dno from csem1 minus select dno from dsem1;

       DNO
----------
        33

SQL> spool off
```

EXPERIMENT NO 06

Implementation of nested queries , string operations, group by clause

in database.

```
SQL> connect

Enter user-name: system

Enter password:

Connected.

SQL> create table dssem1(eid number(10), ename varchar2(10), sal number(10), dno

number(10), post varchar2(10));

Table created.

SQL> select *from dssem1;

no rows selected

SQL> insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post');

Enter value for eid: 10

Enter value for ename: rohit

Enter value for sal: 200000

Enter value for dno: 15

Enter value for post: ceo

old 1: insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post')

new 1: insert into dssem1 values('10', 'rohit', '200000', '15', 'ceo')

1 row created.

SQL> insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post');

Enter value for eid: 25

Enter value for ename: bumrah

Enter value for sal: 100000

Enter value for dno: 20

Enter value for post: coo

old 1: insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post')
```

```
new 1: insert into dssem1 values('25', 'bumrah', '100000', '20', 'coo')
1 row created.
SQL> insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post');
Enter value for eid: 20
Enter value for ename: surya
Enter value for sal: 150000
Enter value for dno: 17
Enter value for post: cap
old 1: insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post')
new 1: insert into dssem1 values('20', 'surya', '150000', '17', 'cap')
1 row created.
SQL> insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post');
Enter value for eid: 11
Enter value for ename: hardik
Enter value for sal: 120000
Enter value for dno: 17
Enter value for post: cap
old 1: insert into dssem1 values('&eid', '&ename', '&sal', '&dno', '&post')
new 1: insert into dssem1 values('11', 'hardik', '120000', '17', 'cap')
1 row created.
SQL> select *from dssem1;
 EID ENAME SAL DNO POST
---------- ---------- ---------- ---------- ----------
 10 rohit 200000 15 ceo
 25 bumrah 100000 20 coo
 20 surya 150000 17 cap
 11 hardik 120000 17 cap
SQL> select *from dssem1 where dno=(select dno from dssem1 where ename='hardik');
```

```
 EID ENAME SAL DNO POST
---------- ---------- ---------- ---------- ----------
 20 surya 150000 17 cap
 11 hardik 120000 17 cap
SQL> select *from dssem1 where sal>(select sal from dssem1 where ename='surya');
 EID ENAME SAL DNO POST
---------- ---------- ---------- ---------- ----------
 10 rohit 200000 15 ceo
SQL> select dno, sum(sal) from dssem1 group by dno having dno=17;
 DNO SUM(SAL)
---------- ----------
 17 270000
SQL> select dno, sum(sal) from dssem1 group by dno having dno in(15,20);
 DNO SUM(SAL)
---------- ----------
 20 100000
 15 200000
SQL> select post, max(sal) from dssem1 group by post;
POST MAX(SAL)
---------- ----------
coo 100000
ceo 200000
cap 150000
SQL> select lower(ename) from dssem1;
LOWER(ENAM
----------
rohit
bumrah
```

```
surya

hardik

SQL> select upper(ename) from dssem1;

UPPER(ENAM

----------

ROHIT

BUMRAH

SURYA

HARDIK

SQL> select upper('captain') from dual;

UPPER('

-------

CAPTAIN

SQL> select lower('captain') from dual;

LOWER('

-------

captain

SQL> select initcap('captain') from dual;

INITCAP

-------

Captain

SQL> select initcap(ename) from dssem1;

INITCAP(EN

----------

Rohit

Bumrah

Surya

Hardik
```

```
SQL> select lpad('kavita', 14, '&') from dual;

LPAD('KAVITA',

--------------

&&&&&&&&kavita

SQL> select rpad('kavita', 14, '&') from dual;

RPAD('KAVITA',

--------------

kavita&&&&&&&&

SQL> select ltrim('kavita', 'k') from dual;

LTRIM

-----

avita

SQL> select rtrim('kavita', 'a') from dual;

RTRIM

-----

kavit

SQL> select translate('rsaggarwal', 'r', 'd') from dual;

TRANSLATE(

----------

dsaggadwal

SQL> select replace('jack and jue', 'j', 'bl') from dual;

REPLACE('JACKA

--------------

black and blue

SQL> select length('jack and jill') from dual;

LENGTH('JACKANDJILL')

---------------------
```

```
SQL> select length(ename) from dssem1;

LENGTH(ENAME)

-------------

 5

 6

 5

 6

SQL> select substr('jackandjill', 5,4) from dual;

SUBS

----

Andj
```

EXPERIMENT NO 07

IMPLEMENTING VARIOUS JOIN OPERATIONS

SQL> create table studentinfo(s_id number(5) not null, s_name varchar(6), dept varchar(4), address varchar(5));

Table created.

SQL> create table accountinfo(s_id number(5) not null, a_name varchar(6), accountno number(4), fees number(5));

Table created.

SQL> insert into studentinfo values('&s_id', '&s_name', '&dept', '&address');

Enter value for s_id: 1

Enter value for s_name: aarav

Enter value for dept: ds

Enter value for address: thane

old 1: insert into studentinfo values('&s_id', '&s_name', '&dept', '&address')

new 1: insert into studentinfo values('1', 'aarav', 'ds', 'thane')

1 row created.

SQL> insert into studentinfo values('&s_id', '&s_name', '&dept', '&address');

Enter value for s_id: 2

Enter value for s_name: om

Enter value for dept: aiml

```
Enter value for address: worli
old   1: insert into studentinfo values('&s_id', '&s_name', '&dept', '&address')
new   1: insert into studentinfo values('2', 'om', 'aiml', 'worli')

1 row created.

SQL> insert into studentinfo values('&s_id', '&s_name', '&dept', '&address');
Enter value for s_id: 3
Enter value for s_name: varun
Enter value for dept: it
Enter value for address: goa
old   1: insert into studentinfo values('&s_id', '&s_name', '&dept', '&address')
new   1: insert into studentinfo values('3', 'varun', 'it', 'goa')

1 row created.


SQL> insert into studentinfo values('&s_id', '&s_name', '&dept', '&address');
Enter value for s_id: 4
Enter value for s_name: sejal
Enter value for dept: cs
Enter value for address: bihar
old   1: insert into studentinfo values('&s_id', '&s_name', '&dept', '&address')
new   1: insert into studentinfo values('4', 'sejal', 'cs', 'bihar')

1 row created.

SQL> insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees');
```

```
Enter value for s_id: 1
Enter value for a_name: aarav
Enter value for accountno: 1234
Enter value for fees: 12345
old   1: insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees')
new   1: insert into accountinfo values('1', 'aarav', '1234', '12345')

1 row created.

SQL> insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees');
Enter value for s_id: 23
Enter value for a_name: tejal
Enter value for accountno: 2345
Enter value for fees: 23456
old   1: insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees')
new   1: insert into accountinfo values('23', 'tejal', '2345', '23456')

1 row created.

SQL> insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees');
Enter value for s_id: 12
Enter value for a_name: yash
Enter value for accountno: 3456
Enter value for fees: 34567
old   1: insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees')
new   1: insert into accountinfo values('12', 'yash', '3456', '34567')

1 row created.
```

```
SQL> insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees');
Enter value for s_id: 13
Enter value for a_name: nidhi
Enter value for accountno: 4567
Enter value for fees: 45678
old   1: insert into accountinfo values('&s_id', '&a_name', '&accountno', '&fees')
new   1: insert into accountinfo values('13', 'nidhi', '4567', '45678')

1 row created.

SQL> select *from studentinfo;

     S_ID S_NAME DEPT ADDRE
---------- ------ ---- -----
         1 aarav  ds   thane
         2 om     aiml worli
         3 varun  it   goa
         4 sejal  cs   bihar

SQL> select *from accountinfo;

     S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---------- ----------
         1 aarav       1234      12345
        23 tejal       2345      23456
        12 yash        3456      34567
        13 nidhi       4567      45678
```

```
SQL> select *from studentinfo, accountinfo;

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
         1 aarav  ds   thane         1 aarav       1234      12345
         1 aarav  ds   thane        23 tejal       2345      23456
         1 aarav  ds   thane        12 yash        3456      34567
         1 aarav  ds   thane        13 nidhi       4567      45678
         2 om     aiml worli         1 aarav       1234      12345
         2 om     aiml worli        23 tejal       2345      23456
         2 om     aiml worli        12 yash        3456      34567
         2 om     aiml worli        13 nidhi       4567      45678
         3 varun  it   goa           1 aarav       1234      12345
         3 varun  it   goa          23 tejal       2345      23456
         3 varun  it   goa          12 yash        3456      34567

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
         3 varun  it   goa          13 nidhi       4567      45678
         4 sejal  cs   bihar         1 aarav       1234      12345
         4 sejal  cs   bihar        23 tejal       2345      23456
         4 sejal  cs   bihar        12 yash        3456      34567
         4 sejal  cs   bihar        13 nidhi       4567      45678

16 rows selected.
```

```
SQL> select studentinfo.s_id, s_name, dept, accountno, fees from studentinfo, accountinfo
where studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT  ACCOUNTNO       FEES
---------- ------ ---- ---------- ----------
         1 aarav  ds        1234      12345

SQL> select studentinfo.s_id, s_name, dept from studentinfo inner join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT
---------- ------ ----
         1 aarav  ds

SQL> select studentinfo.s_id, s_name, dept from studentinfo left join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT
---------- ------ ----
         1 aarav  ds
         4 sejal  cs
         3 varun  it
         2 om     aiml

SQL> select studentinfo.s_id, s_name, dept from studentinfo right join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT
---------- ------ ----
```

```
        1 aarav  ds
```

```
SQL> select studentinfo.s_id, s_name, dept from studentinfo full join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT
---------- ------ ----
        1 aarav  ds
        4 sejal  cs
        3 varun  it
        2 om     aiml

7 rows selected.
```

```
SQL> select *from studentinfo right outer join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
        1 aarav  ds   thane         1 aarav       1234      12345
                                  13 nidhi        4567      45678
                                  23 tejal        2345      23456
                                  12 yash         3456      34567
```

```
SQL> select *from studentinfo left outer join accountinfo on
studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
         1 aarav  ds   thane         1 aarav       1234      12345
         4 sejal  cs   bihar
         3 varun  it   goa
         2 om     aiml worli

SQL> select *from studentinfo inner join accountinfo on studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
         1 aarav  ds   thane         1 aarav       1234      12345

SQL> select *from studentinfo join accountinfo on studentinfo.s_id=accountinfo.s_id;

     S_ID S_NAME DEPT ADDRE       S_ID A_NAME  ACCOUNTNO       FEES
---------- ------ ---- ----- ---------- ------ ---------- ----------
         1 aarav  ds   thane         1 aarav       1234      12345
```

Experiment No: 08

Implementation of Transaction Control Language (TCL):

CODE:

```
SQL> create table stud_info (Sid number (10), Sname varchar2 (10), Address varchar2 (10),
dept_no number

(10));

Table created.

SQL> insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no');

Enter value for sid: 1

Enter value for sname: Hamza

Enter value for address: Lyari

Enter value for dept_no: 45

old 1: insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no')

new 1: insert into stud_info values ('1', 'Hamza', 'Lyari', '45')

1 row created.

SQL> insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no');

Enter value for sid: 3

Enter value for sname: Aalam

Enter value for address: Doodh Soda

Enter value for dept_no: 18

old 1: insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no')

new 1: insert into stud_info values ('3', 'Aalam', 'Doodh Soda', '18')

1 row created.

SQL> insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no');

Enter value for sid: 40
```

```
Enter value for sname: Rizwan

Enter value for address: Pappu

Enter value for dept_no: 63

old 1: insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no')

new 1: insert into stud_info values ('40', 'Rizwan', 'Pappu', '63')

1 row created.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Pappu 63

SQL> insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no');

Enter value for sid: 2

Enter value for sname: Jameel

Enter value for address: Karachi

Enter value for dept_no: 7

old 1: insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no')

new 1: insert into stud_info values ('2', 'Jameel', 'Karachi', '7')

1 row created.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Pappu 63

2 Jameel Karachi 7

SQL> savepoint o1;
```

```
Savepoint created.

SQL> update stud_info set sid=20 where dept_no=7;

1 row updated.

SQL> insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no');

Enter value for sid: 30

Enter value for sname: SP

Enter value for address: LTF

Enter value for dept_no: 93

old 1: insert into stud_info values ('&Sid', '&Sname', '&Address', '&dept_no')

new 1: insert into stud_info values ('30', 'SP', 'LTF', '93')

1 row created.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Pappu 63

20 Jameel Karachi 7

30 SP LTF 93

SQL> delete from stud_info where address='LTF';

1 row deleted.

SQL> savepoint o2;

Savepoint created.

SQL> update stud_info set address='Arshad' where sid=40;

1 row updated.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------
```

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Arshad 63

20 Jameel Karachi 7

SQL> rollback to o2;

Rollback complete.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Pappu 63

20 Jameel Karachi 7

SQL> rollback to o1;

Rollback complete.

SQL> select *from stud_info;

SID SNAME ADDRESS DEPT_NO

---------- ---------- ---------- ----------

1 Hamza Lyari 45

3 Aalam Doodh Soda 18

40 Rizwan Pappu 63

2 Jameel Karachi 7

SQL> commit;

Commit complete.

SQL> spool off

Experiment 9

SQL> create view stud_v1 as select sid, sname from student;

View created. SQL> select *from stud_v1;

SID SNAME

---------- ---------- 20 Ananya

30 Mayank

40 Bharati

SQL> create or replace view stud_v1 as select sid, sname, dept from student;

View created. SQL> replace view stud_v2 as select sid, sname, dept from student;

SQL> select *from stud_v1;

SID SNAME DEPT

---------- ---------- ---------- 20 Ananya 10000

30 Mayank 40000

40 Bharati 50000

SQL> update stud_v1 set sname='Akanksha' where sid=40;

1 row updated. SQL> select *from stud_v1;

SID SNAME DEPT

---------- ---------- ---------- 20 Ananya 10000

30 Mayank 40000

40 Akanksha 50000

SQL> select *from student;

SID SNAME DEPT

---------- ---------- ---------- 20 Ananya 10000

30 Mayank 40000

40 Akanksha 50000

SQL> insert into stud_v1 values(35, 'Dhruv', 'FINANVE');

1 row created. SQL> select *from stud_v1;

SID SNAME DEPT

---------- ---------- ---------- 35 Dhruv FINANVE

20 Ananya 10000

30 Mayank 40000

40 Akanksha 50000

SQL> select *from student;

SID SNAME DEPT

---------- ---------- ---------- 35 Dhruv FINANVE

20 Ananya 10000

30 Mayank 40000

40 Akanksha 50000

SQL> create table lib(sid number(10), subject varchar2(10), fine number(10));

Table created. SQL> insert into lib values(35, 'python', 200);

1 row created. SQL> insert into lib values(25, 'c++', 100);

1 row created. SQL> insert into lib values(35, 'c++', 100);

1 row created. SQL> insert into lib values(45, 'java', 100);

1 row created. SQL> insert into lib values(25, 'dbms', 100);

1 row created. SQL> select *from lib;

SID SUBJECT FINE

---------- ---------- ---------- 35 python 200

25 c++ 100

35 c++ 100

45 java 100

25 dbms 100

SQL> create or replace view lib1 as select sid, fine from lib;

View created. SQL> select *from lib1;

SID FINE

---------- ---------- 35 200

25 100

35 100

45 100

25 100

SQL> create or replace view lib2 as select sid, sum(fine) total_fine from lib group by sid;

View created. SQL> select *from lib2;

SID TOTAL_FINE

---------- ---------- 25 200

35 300

45 100

SQL> create or replace trigger lib_t1

2 before update on lib

3 begin

4 dbms_output.put_line('library record updated');

5 end;

6 /

Trigger created. SQL> update lib

2 set fine=fine*2

3 where fine<200;

4 rows updated. SQL> select *from lib;

SID SUBJECT FINE

---------- ---------- ---------- 35 python 200

25 c++ 200

35 c++ 200

45 java 200

25 dbms 200

Experiment 10

```
SQL> create table studinfo12(sid number(5) NOT NULL, name varchar(10));

Table created.

SQL> desc studinfo12;

Name Null? Type

---------- -------- ----------

SID NOT NULL NUMBER(5)

NAME VARCHAR2(10)

SQL> insert into studinfo12 values('&sid', '&name');

Enter value for sid: 1

Enter value for name: ak

old 1: insert into studinfo12 values('&sid', '&name')

new 1: insert into studinfo12 values('1', 'ak')

1 row created.

SQL> insert into studinfo12 values('&sid', '&name');

Enter value for sid:

Enter value for name: varsha

old 1: insert into studinfo12 values('&sid', '&name')

new 1: insert into studinfo12 values('', 'varsha')

insert into studinfo12 values('', 'varsha')

 *

ERROR at line 1:

ORA-01400: cannot insert NULL into ("SYSTEM"."STUDINFO12"."SID")

SQL> create table studinfo11(sid number(5) UNIQUE, name varchar(10));

Table created.

SQL> desc studinfo11;

Name Null? Type

---------- -------- ----------
```

```
SID NUMBER(5)

NAME VARCHAR2(10)

SQL> insert into studinfo11 values('&sid', '&name');

Enter value for sid: 1

Enter value for name: ak

old 1: insert into studinfo11 values('&sid', '&name')

new 1: insert into studinfo11 values('1', 'ak')

1 row created.

SQL> insert into studinfo11 values('&sid', '&name');

Enter value for sid: 1

Enter value for name: nidhi

old 1: insert into studinfo11 values('&sid', '&name')

new 1: insert into studinfo11 values('1', 'nidhi')

insert into studinfo11 values('1', 'nidhi')

*

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.SYS_C007005) violated

SQL> create table studinfo13(sid number(5), name varchar(10), dept varchar(5), CHECK(dept

IN('ds', 'it', 'cs')));

Table created.

SQL> insert into studinfo13 values('&sid', '&name', '&dept');

Enter value for sid: 1

Enter value for name: ak

Enter value for dept: ds

old 1: insert into studinfo13 values('&sid', '&name', '&dept')

new 1: insert into studinfo13 values('1', 'ak', 'ds')

1 row created.

SQL> 2
```

```
SP2-0226: Invalid line number

SQL> insert into studinfo13 values('&sid', '&name', '&dept');

Enter value for sid: 2

Enter value for name: varsha

Enter value for dept: aiml

old 1: insert into studinfo13 values('&sid', '&name', '&dept')

new 1: insert into studinfo13 values('2', 'varsha', 'aiml')

insert into studinfo13 values('2', 'varsha', 'aiml')

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007006) violated

SQL> create table faculty(fid number(5) primary key, fname varchar(10));

Table created.

SQL> desc faculty;

Name Null? Type

----------------------------------------- -------- ----------------------------

FID NOT NULL NUMBER(5)

FNAME VARCHAR2(10)

SQL> create table subject10(subid number(5) primary key, subname varchar(10), fid
number(5),

foreign key(fid) references faculty);

Table created.

SQL> desc subject10;

Name Null? Type

----------------------------------------- -------- ----------------------------

SUBID NOT NULL NUMBER(5)

SUBNAME VARCHAR2(10)

FID NUMBER(5)
```